



ORDER: 240315-2038
TEST: U240315-2038-1
CLIENT REF: ord_jM7Vlb1
PATIENT: Hannah Munguia
ID: MUNGUIA-H-00018
SEX: Female
AGE: 28 DOB: 08/18/1995

CLIENT #: 47938
Rupa Health
177 Townsend Street Unit 528
San Francisco, CA 94107 U.S.A.

Toxic Metals; urine

TOXIC METALS					
		RESULT µg/g Creat	REFERENCE INTERVAL	WITHIN REFERENCE	OUTSIDE REFERENCE
Aluminum	(Al)	4.4	< 25		
Antimony	(Sb)	0.12	< 0.18		
Arsenic	(As)	21	< 50		
Barium	(Ba)	1.0	< 5		
Beryllium	(Be)	<dl	< 0.01		
Bismuth	(Bi)	19	< 1		
Cadmium	(Cd)	0.25	< 0.9		
Cesium	(Cs)	4.2	< 10		
Gadolinium	(Gd)	<dl	< 0.8		
Lead	(Pb)	4.5	< 1.2		
Mercury	(Hg)	10	< 1.3		
Nickel	(Ni)	3.4	< 5		
Palladium	(Pd)	<dl	< 0.3		
Platinum	(Pt)	<dl	< 0.1		
Tellurium	(Te)	<dl	< 0.5		
Thallium	(Tl)	0.37	< 0.5		
Thorium	(Th)	<dl	< 0.02		
Tin	(Sn)	0.61	< 5		
Tungsten	(W)	0.13	< 0.4		
Uranium	(U)	<dl	< 0.03		

URINE CREATININE					
	RESULT mg/dL	REFERENCE INTERVAL	-2SD	-1SD	MEAN
Creatinine	49.8	30 – 225			

SPECIMEN DATA		
Comments:		
Date Collected: 03/13/2024	Provoking Agent: DMSA 2000MG	pH upon receipt: Acceptable
Date Received: 03/15/2024	Provocation: Post Provocative	
Date Reported: 03/18/2024	Collection Period: 6 hours	
Methodology: ICP-MS QQQ, Creatinine by Jaffe Reaction		

< dl: less than detection limit

Results are creatinine corrected to account for urine dilution variations. Reference intervals are based upon NHANES (cdc.gov/nhanes) data if available, and are representative of a large population cohort under non-provoked conditions. Chelation (provocation) agents can increase urinary excretion of metals/elements.

Introduction

This analysis of urinary elements was performed by ICP-Mass Spectroscopy following acid digestion of the specimen. Urine element analysis is intended primarily for: diagnostic assessment of toxic element status, monitoring detoxification therapy, and identifying or quantifying renal wasting conditions. It is difficult and problematic to use urinary elements analysis to assess nutritional status or adequacy for essential elements. Blood, cell, and other elemental assimilation and retention parameters are better indicators of nutritional status.

- **24 Hour Collections**

"Essential and other" elements are reported as mg/24 h; mg element/urine volume (L) is equivalent to ppm. "Potentially Toxic Elements" are reported as µg/24 h; µg element/urine volume (L) is equivalent to ppb.

- **Timed Samples (< 24 hour collections)**

All "Potentially Toxic Elements" are reported as µg/g creatinine; all other elements are reported as µg/mg creatinine. Normalization per creatinine reduces the potentially great margin of error which can be introduced by variation in the sample volume. It should be noted, however, that creatinine excretion can vary significantly within an individual over the course of a day.

If one intends to utilize urinary elements analysis to assess nutritional status or renal wasting of essential elements, it is recommended that unprovoked urine samples be collected for a complete 24 hour period. For provocation (challenge) tests for potentially toxic elements, shorter timed collections can be utilized, based upon the pharmacokinetics of the specific chelating agent. When using EDTA, DMPS or DMSA, urine collections up to 12 hours are sufficient to recover greater than 90% of the mobilized metals. Specifically, we recommend collection times of: 9 - 12 hours post intravenous EDTA, 6 hours post intravenous or oral DMPS and, 6 hours post oral bolus administration of DMSA. What ever collection time is selected by the physician, it is important to maintain consistency for subsequent testing for a given patient.

If an essential element is sufficiently abnormal per urine measurement, a descriptive text is included with the report. Because renal excretion is a minor route of excretion for some elements, (Cu, Fe, Mn Zn), urinary excretion may not influence or reflect body stores. Also, renal excretion for many elements reflects homeostasis and the loss of quantities that may be at higher dietary levels than is needed temporarily. For these reasons, descriptive texts are provided for specific elements when deviations are clinically significant. For potentially toxic elements, a descriptive text is provided whenever levels are measured to be higher than expected. If no descriptive texts follow this introduction, then all essential element levels are within acceptable range and all potentially toxic elements are within expected limits.

Reference intervals and corresponding graphs shown in this report are representative of a healthy population under non-provoked conditions. Descriptive texts appear in this report on the basis of measured results and correspond to non-challenge, non-provoked conditions.

Chelation (provocation) agents can increase urinary excretion of metals/elements. Provoked reference intervals have not been established therefore non-provoked reference intervals shown are not recommended for comparison purposes with provoked test results. Provoked results can be compared with non-provoked results (not reference intervals) to assess body burden of metals and to distinguish between transient exposure and net retention of metals. Provoked results can also be compared to previous provoked results to monitor therapies implemented by the treating physician. Additionally, Ca-EDTA provoked results can be used to calculate the EDTA/Lead Excretion Ratio (LER) in patients with elevated blood levels.

CAUTION: Even the most sensitive instruments have some detection limit below which a measurement cannot be made reliably. Any value below the method detection limit is simply reported as "< dl." If an individual excretes an abnormally high volume of urine, urinary components are likely to be extremely dilute. It is possible for an individual to excrete a relatively large amount of an element per day that is so diluted by the large urine volume that the value measured is near the dl. This cannot automatically be assumed to be within the reference range.

This analysis of urinary metals was performed by ICP-Mass Spectroscopy. Urine metal analysis is traditionally used for evaluation of very recent or ongoing exposure to potentially toxic metals. The urinary excretion of certain metals is known to be increased (provoked) to a variable extent after administration of specific chelating agents. Reference values and corresponding graphs are representative of a healthy population under non-provoked conditions; reference values have not been established for provoked urine samples. Reference values are age and sex specific.

For timed, random or first morning urine collections, metals are reported as µg/ gram creatinine. Normalization per creatinine reduces the potentially great margin of error that can be introduced by variation in the sample volume (concentration). It should be noted that creatinine excretion for an individual may vary to some extent over the course of a day, and from day to day. For 24 hour (h) urine collections elements are reported as µg/24 h. Results are also reported as ug element/gram creatinine to ensure clinically useful information in the event that an inaccurate 24 h urine volume was reported to the laboratory.

Descriptive texts appear in this report if detected levels of specific elements are abnormally high by comparison to the unprovoked reference values. If no descriptive texts follow this introduction, potentially toxic metals are within reference limits.

Bismuth High

This individual's urine bismuth is higher than expected. Urine is the principal mode for excretion of absorbed bismuth. This element is considered to be only slightly toxic with ingestion of gram quantities necessary before signs of toxicity occur. Only between 5 and 10% of orally ingested, soluble bismuth salts are absorbed into the blood.

Bismuth is a byproduct of lead and copper ore refining. Bismuth has therapeutic uses with antimicrobial, anti-secretory and anti-inflammatory actions. Bismuth subsalicylate ("Pepto-Bismol") hydrolyzes in the stomach to salicylic acid and insoluble bismuth; it can be effective in halting traveler's diarrhea. Historically, bismuth was used to treat syphilis. Bismuth is used commercially in low-melting-point alloys and solders and is commonly in "automatic" sprinkler heads for in-building fire protection. Bismuth often is a component of: pigments, paints, glazes for ceramics, glass, and some semiconductor materials. Some cosmetics including lipstick may contain bismuth oxides as a pigment (pearlescent white). Dry cell battery electrodes (cathode) may contain bismuth.

At sub-gram quantities, no toxic effects are documented for bismuth. Also, the existence of health problems due to environmental pollution by bismuth is not documented (Tsalev p. 101, 1983). Early physiological signs of bismuth excess may include: constipation or bowel irregularity, foul breath, skin pigmentation changes, and gum pigmentation (blue-black) with stomatitis.

Laboratory tests that help to assess bismuth status are whole blood and hair element analyses. Some increase in urine bismuth may follow administration of dithiol chelators (DMPS, DMSA). Bismuth has a very high affinity for sulfhydryl groups.

Lead High

This individual's urine lead (Pb) is higher than expected which means that Pb exposure is higher than that of the general population. A percentage of assimilated Pb is excreted in urine. Therefore the urine Pb level reflects recent or ongoing exposure to Pb and the degree of excretion or endogenous detoxification processes.

Sources of Pb include: old lead-based paints, batteries, industrial smelting and alloying, some types of solders, Ayurvedic herbs, some toys and products from China and Mexico, glazes on (foreign) ceramics, leaded (anti-knock compound) fuels, bullets and fishing sinkers, artist paints with Pb pigments, and leaded joints in municipal water systems. Most Pb contamination occurs via oral ingestion of contaminated food or water or by children mouthing or eating Pb-containing substances. The degree of absorption of oral Pb depends upon stomach contents (empty stomach increases uptake) and upon the essential element intake and Pb status. Deficiency of zinc, calcium or iron increases Pb uptake. Transdermal exposure is significant for Pb-acetate (hair blackening products). Inhalation has decreased significantly with almost universal use of non-leaded automobile fuel.

Lead accumulates extensively in bone and can inhibit formation of heme and hemoglobin in erythroid precursor cells. Bone Pb is released to soft tissues with bone remodeling that can be accelerated with growth, menopausal hormonal changes, osteoporosis, or skeletal injury. Low levels of Pb may cause impaired vitamin D metabolism, decreased nerve conduction, and developmental problems for children including: decreased IQ, hearing impairment, delayed growth, behavior disorders, and decreased glomerular function. Transplacental transfer of Pb to the fetus can occur at very low Pb concentrations in the body. At relatively low levels, Pb can participate in synergistic toxicity with other toxic elements (e.g. cadmium, mercury).

Excessive Pb exposure can be assessed by comparing urine Pb levels before and after provocation with Ca-EDTA (iv) or oral DMSA. Urine Pb is higher post-provocation to some extent in almost everyone. Whole blood analysis reflects only recent ongoing exposure and does not correlate well with total body retention of Pb. However, elevated blood Pb is the standard of care for diagnosis of Pb poisoning (toxicity).

Mercury High

This individual's urine mercury (Hg) is higher than expected but may not be sufficiently high to be associated with overt pathophysiological effects. Symptomatology depends on many factors: the chemical form of Hg, its accumulation in specific tissues, presence of other toxicants, presence of disease that depletes glutathione or inactivates lymphocytes or is immunosuppressive, and the concentration of protective nutrients, (e.g. zinc, selenium).

Early signs of excessive Hg exposure include: decreased senses of touch, hearing, vision and taste, metallic taste in mouth, fatigue or lack of physical endurance, and increased salivation. Symptoms may progress with moderate or chronic exposure to include: anorexia, numbness and paresthesias, headaches, hypertension, irritability and excitability, and immune suppression/ dysregulation. Advanced disease processes from excessive Hg assimilation include: tremors and incoordination, anemia, psychoses, manic behaviors, possibly autoimmune disorders and renal dysfunction or failure. Note that in Hg exposure of long duration, renal excretion of Hg (and normal metabolites) may become impaired, and the urine level of Hg might be only mildly elevated or not elevated at all due to renal failure.

Mercury is used in: dental amalgams (50% by weight), explosive detonators; some vaccines, pure liquid form in thermometers, barometers, and laboratory equipment; batteries and electrodes, some medications and Ayurvedic herbs, fungicides and pesticides, and in the paper industry. The fungicide/pesticide use of mercury has declined due to environmental concerns, but Hg residues persist in the environment. Emissions from coal-fired power plants and hospital/municipal incinerators are significant sources of mercury pollution.

Methylmercury, the most common, organic form of Hg, occurs by methylation of inorganic Hg in aquatic biota or sediments (both freshwater and ocean sediments). Methylmercury accumulates in aquatic animals and fish and is concentrated up the food chain reaching highest concentrations in large fish and predatory birds. Except for fish, the human intake of dietary mercury is negligible unless the food is contaminated with one of the previously listed forms/sources. Daily ingestion of fish can result in the assimilation of 1 to 10 micrograms of mercury/day.

Depending upon the extent of cumulative Hg exposure, elevated levels of urine Hg may occur after administration of DMPS, DMSA or D-penicillamine. Blood and especially red blood cell elemental analyses are useful for assessing recent or ongoing exposure to organic (methyl) Hg.